Statistical Computing

Bootstrap and Jackknife

Spring 2026

Outline

The Bootstrap

Bootstrap: overview

Bootstrap methods

Bootstrap methods are a class of *nonparametric Monte Carlo* methods that estimate the distribution of a population by *resampling*.

- Resampling methods treat an observed sample as a *finite population*, and random samples are generated (resampled) from it to estimate population characteristics and make inferences.
- Bootstrap methods are often used when the distribution of the target population is not specified; the sample is the only information available.
- The term "bootstrap" here refers to *nonparametric* bootstrap (sampling from the data); Monte Carlo methods that sample from a fully specified distribution are called *parametric* bootstrap.

Resampling and the empirical distribution

- Let $\boldsymbol{x} = (x_1, \ldots, x_n)$ be an observed random sample from a distribution with cdf $F(x)$.

Bootstrap resampling

If X^* is selected at random from $\boldsymbol{x}$, then

$$\mathbb{P}(X^* = x_i) = \frac{1}{n}, \quad i = 1, \ldots, n,$$

and a bootstrap sample $X_1^*, \ldots, X_n^*$ is generated by sampling *with replacement* from $\boldsymbol{x}$. The variables X_i^* are iid, uniformly distributed on $\{x_1, \ldots, x_n\}$.

Empirical distribution function (recall)

Definition

An estimator of $F(x) = \mathbb{P}(X \leq x)$ is the proportion of sample points that fall in $(-\infty, x]$.

The empirical cdf (ecdf) is

$$F_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i \leq x\}.$$

Piecewise form

Let $x_{(1)} \leq \cdots \leq x_{(n)}$ be the ordered sample. Then

$$F_n(x) = \begin{cases} 0, & x < x_{(1)}, \\ \dfrac{i}{n}, & x_{(i)} \leq x < x_{(i+1)}, \quad i = 1, \ldots, n-1, \\ 1, & x_{(n)} \leq x. \end{cases}$$

Two approximations in bootstrap

$F_n(x)$ is itself the distribution function of a random variable uniformly distributed on $\{x_1, \ldots, x_n\}$. In bootstrap, there are two approximations:

- $F_n(x)$ approximates $F(x)$.
- The ecdf of the bootstrap replicates, $\widehat{F}^*$, approximates $F_n(x)$.

$$F \longrightarrow X \longrightarrow F_n, \qquad F_n \longrightarrow X^* \longrightarrow \widehat{F}^*.$$

- Resampling from $\boldsymbol{x}$ is equivalent to generating random samples from $F_n(x)$.

Bootstrap sampling procedure

Let θ be the parameter of interest and $\hat{\theta}$ an estimator.

1. For each replicate $b = 1, \ldots, B$:
   1.1 Generate $\boldsymbol{x}^{*(b)}$ by sampling with replacement from $\boldsymbol{x}$.
   1.2 Compute $\hat{\theta}^{(b)}$ from $\boldsymbol{x}^{*(b)}$.
2. The bootstrap estimate of the distribution of $\hat{\theta}$ is the empirical distribution of $\hat{\theta}^{(1)}, \ldots, \hat{\theta}^{(B)}$.

Example: F_n and bootstrap samples

- Observed sample:

$$\boldsymbol{x} = \{2, 2, 1, 1, 5, 4, 4, 3, 1, 2\}.$$

- Resampling from $\boldsymbol{x}$ selects $1, 2, 3, 4, 5$ with probabilities $0.3, 0.3, 0.1, 0.2, 0.1$, respectively.
- Therefore the cdf of a randomly selected replicate equals the ecdf $F_n(x)$:

$$F_n(x) = \begin{cases} 0, & x < 1, \\ 0.3, & 1 \leq x < 2, \\ 0.6, & 2 \leq x < 3, \\ 0.7, & 3 \leq x < 4, \\ 0.9, & 4 \leq x < 5, \\ 1, & x \geq 5. \end{cases}$$

- Note: if F_n is not close to F, the distribution of replicates will not be close to F.

Bootstrap estimate of standard error

Bootstrap standard error

The bootstrap estimate of standard error of an estimator $\hat{\theta}$ is the sample standard deviation of the bootstrap replicates $\hat{\theta}^{(1)}, \ldots, \hat{\theta}^{(B)}$:

$$\widehat{\mathrm{se}}_B(\hat{\theta}^*) = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}\left(\hat{\theta}^{(b)} - \bar{\theta}^*\right)^2}, \qquad \bar{\theta}^* = \frac{1}{B}\sum_{b=1}^{B}\hat{\theta}^{(b)}.$$

- The number of replicates needed for good estimates of standard error is typically not large: $B = 50$ is usually large enough, and rarely is $B > 200$ necessary.
- (Much larger B will be needed for confidence interval estimation.)

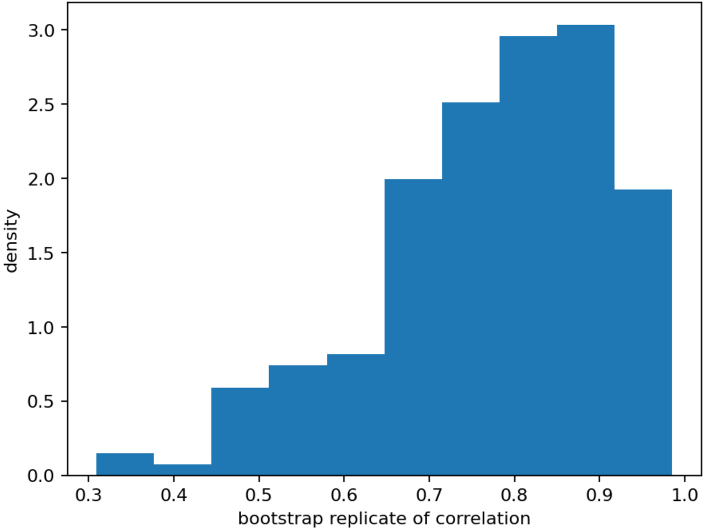
Example: law school data (correlation)

- Data: LSAT and GPA for $n = 15$ law schools (random sample from a universe of 82 law schools).
- Goal: estimate the correlation between LSAT and GPA, and compute the bootstrap estimate of the standard error of the sample correlation.

$$\text{LSAT: } 576, 635, 558, 578, 666, 580, 555, 661, 651, 605, 653, 575, 545, 572, 594$$

$$\text{GPA: } 339, 330, 281, 303, 344, 307, 300, 343, 336, 313, 312, 274, 276, 288, 296$$

Figure: histogram of bootstrap replicates



Bootstrap estimation of bias

Bias

If $\hat{\theta}$ is an unbiased estimator of θ, then $\mathbb{E}[\hat{\theta}] = \theta$. The bias of an estimator $\hat{\theta}$ for θ is

$$\mathrm{bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta} - \theta] = \mathbb{E}[\hat{\theta}] - \theta.$$

- Every statistic is an unbiased estimator of its expected value. In particular, the sample mean is an unbiased estimator of the population mean.
- Example of a biased estimator: the MLE of variance,

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad \mathbb{E}[\hat{\sigma}^2] = \left(1 - \frac{1}{n}\right)\sigma^2,$$

so $\hat{\sigma}^2$ underestimates σ^2 and the bias is $-\sigma^2/n$.

Bootstrap estimate of bias

- Bootstrap estimation of bias uses bootstrap replicates of $\hat{\theta}$ to estimate the sampling distribution of $\hat{\theta}$.
- For the finite population $\boldsymbol{x} = (x_1, \ldots, x_n)$, the parameter is $\hat{\theta}(\boldsymbol{x})$ and we have B iid bootstrap replicates $\hat{\theta}^{(b)}$.

Bootstrap bias estimate

Let

$$\bar{\theta}^* = \frac{1}{B}\sum_{b=1}^{B} \hat{\theta}^{(b)}, \qquad \hat{\theta} = \hat{\theta}(\boldsymbol{x}).$$

The bootstrap estimate of bias is

$$\widehat{\mathrm{bias}}(\hat{\theta}) = \bar{\theta}^* - \hat{\theta}.$$

(Here, θ is replaced by $\hat{\theta}$ because bootstrap samples F_n in place of F.)

Example: bias of the sample correlation (law school data)

- Using the law school data from the previous example, compute the bootstrap estimate of bias in the sample correlation.
- Use $B = 2000$ bootstrap replicates (larger B is used for estimating bias).

Rule of thumb: if $|\mathrm{bias}|/\mathrm{se} \leq 0.25$, it is not usually necessary to adjust for bias. Here the bias is small relative to the standard error (cv < 0.1), so adjustment is not necessary.

The Jackknife

The jackknife: overview

- The jackknife is another resampling method:
- The jackknife is like a "leave-one-out" type of cross-validation.

Jackknife samples and replicates

Let $\boldsymbol{x} = (x_1, \ldots, x_n)$ be an observed random sample. Define the *i*th jackknife sample by leaving out x_i:

$$\boldsymbol{x}^{(i)} = (x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n).$$

If $\hat{\theta} = T_n(\boldsymbol{x})$, define the *i*th jackknife replicate

$$\hat{\theta}^{(i)} = T_{n-1}\left(\boldsymbol{x}^{(i)}\right), \qquad i = 1, \ldots, n.$$

Jackknife estimate of bias

Bias estimate

Let

$$\hat{\theta}^{(\cdot)} = \frac{1}{n}\sum_{i=1}^{n}\hat{\theta}^{(i)}, \qquad \hat{\theta} = \hat{\theta}(\boldsymbol{x}).$$

The jackknife estimate of bias is

$$\widehat{\text{bias}}_{\text{jack}} = (n-1)\left(\hat{\theta}^{(\cdot)} - \hat{\theta}\right).$$

- The jackknife requires only n replications to estimate bias.
- The bootstrap estimate of bias typically requires several hundred replicates.

Why the factor $(n-1)$ appears

- Consider the case where θ is the population variance σ_X^2. The plug-in estimate is

$$\hat{\theta} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad \mathrm{bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta} - \sigma_X^2] = -\frac{\sigma_X^2}{n}.$$

- Each jackknife replicate uses sample size $n-1$, so its bias is $-\sigma_X^2/(n-1)$.
- Therefore

$$\mathbb{E}\left[\hat{\theta}^{(i)} - \hat{\theta}\right] = \mathrm{bias}(\hat{\theta}^{(i)}) - \mathrm{bias}(\hat{\theta}) = -\frac{\sigma_X^2}{n(n-1)} = \frac{\mathrm{bias}(\hat{\theta})}{n-1},$$

so multiplying the leave-one-out difference by $(n-1)$ yields the correct bias for this plug-in estimator.

Example: jackknife bias (patch data)

- Data: hormone measurements for $n = 8$ subjects after wearing a medical patch.
- Compute the jackknife estimate of bias for the patch data (ratio statistic)

$$\hat{\theta} = \frac{\bar{y}}{\bar{z}}.$$

subject	placebo	oldpatch	newpatch	z	y
1	9243	17649	16449	8406	-1200
2	9671	12013	14614	2342	2601
3	11792	19979	17274	8187	-2705
4	13357	21816	23798	8459	1982
5	9055	13850	12560	4795	-1290
6	6290	9806	10157	3516	351
7	12412	17208	16570	4796	-638
8	18806	29044	26325	10238	-2719

Jackknife estimate of standard error

Standard error estimate

A jackknife estimate of standard error is

$$\widehat{\mathrm{se}}_{\mathrm{jack}} = \sqrt{\frac{n-1}{n}\sum_{i=1}^{n}\left(\hat{\theta}^{(i)} - \hat{\theta}^{(\cdot)}\right)^2}.$$

- The factor $(n-1)/n$ makes $\widehat{\mathrm{se}}_{\mathrm{jack}}$ an unbiased estimator of the standard error of the sample mean.

When the jackknife fails

- The jackknife can fail when the statistic $\hat{\theta}$ is not "smooth."
- Smoothness means that small changes in the data correspond to small changes in the statistic.
- The median is an example of a statistic that is not smooth.

Details and results (as shown in the text)

- The sample:

$$x: \ 29, \ 79, \ 41, \ 86, \ 91, \ 5, \ 50, \ 83, \ 51, \ 42$$

- Jackknife medians:

$$51, \ 50, \ 51, \ 50, \ 50, \ 51, \ 51, \ 50, \ 50, \ 51$$

- Jackknife estimate of se: 1.5.
- Bootstrap estimate of se (from replicates of the median): 13.69387.
- The jackknife fails here because the median is not smooth.

Bootstrap Confidence Intervals

Bootstrap confidence intervals

- In this section, several approaches to obtaining approximate confidence intervals for the target parameter in a bootstrap are discussed.
- Methods include:
  - the standard normal bootstrap confidence interval,
  - the basic bootstrap confidence interval,
  - the bootstrap percentile confidence interval.

Standard normal bootstrap confidence interval

If $\hat{\theta}$ is approximately normal and unbiased for θ, then

$$Z = \frac{\hat{\theta} - \theta}{\mathrm{se}(\hat{\theta})} \dot{\sim} N(0,1),$$

giving the approximate $100(1-\alpha)\%$ confidence interval

Z-interval

$$\hat{\theta} \pm z_{\alpha/2}\,\mathrm{se}(\hat{\theta}), \qquad z_{\alpha/2} = \Phi^{-1}(1-\alpha/2).$$

In practice $\mathrm{se}(\hat{\theta})$ is unknown. The *bootstrap normal CI* plugs in the bootstrap SE estimate:

$$\hat{\theta} \pm z_{\alpha/2}\,\widehat{\mathrm{se}}_B(\hat{\theta}^*),$$

where $\widehat{\mathrm{se}}_B$ is the sample standard deviation of the bootstrap replicates $\hat{\theta}^{(1)}, \ldots, \hat{\theta}^{(B)}$.

Assumptions and remarks (normal CI)

- This interval is easy to compute, but it relies on assumptions:
  - the distribution of $\hat{\theta}$ is approximately normal;
  - $\hat{\theta}$ is unbiased for θ.
- These assumptions do not always hold — especially when n is small or $\hat{\theta}$ is not a simple mean.
- The basic and percentile intervals (next) do not require normality of $\hat{\theta}$.

Basic bootstrap confidence interval

- The basic bootstrap confidence interval (pivotal CI) transforms the distribution of the replicates by subtracting the observed statistic.
- The quantiles of the transformed sample $\hat{\theta}^* - \hat{\theta}$ are used to determine confidence limits.

Basic (pivotal) CI

Let $\hat{\theta}^*_q$ denote the q sample quantile of the bootstrap replicates $\hat{\theta}^*$. Then the $100(1-\alpha)\%$ basic bootstrap confidence limits are

$$\left(2\hat{\theta} - \hat{\theta}^*_{1-\alpha/2},\ 2\hat{\theta} - \hat{\theta}^*_{\alpha/2}\right).$$

Derivation idea (basic CI)

- Observe a simple random sample $x_1, \ldots, x_n$, and compute $\hat{\theta} = \hat{\theta}(x_1, \ldots, x_n)$.
- Generate B replicates $\hat{\theta}^*$ from an ordinary bootstrap.
- For a symmetric $100(1-\alpha)\%$ confidence interval (L, U) for θ based on bootstrap replicates, require

$$\Pr(L > \theta) = \Pr(U < \theta) = \alpha/2.$$

Derivation steps (basic CI)

- For the lower limit:

$$\alpha/2 = \Pr(L > \theta) = \Pr(L - \hat{\theta} > \theta - \hat{\theta}) = \Pr(\hat{\theta} - \theta > \hat{\theta} - L).$$

- Hence $1 - \alpha/2 = \Pr(\hat{\theta} - \theta \leq \hat{\theta} - L)$, so $\hat{\theta} - L$ is the $(1 - \alpha/2)$ quantile of $\hat{\theta} - \theta$.
- These exact percentiles are unknown, but can be estimated from the bootstrap replicates:

$$\hat{\theta}^*_{1-\alpha/2} - \hat{\theta} \approx \text{the } (1 - \alpha/2) \text{ quantile of } (\hat{\theta} - \theta).$$

- Set $\hat{\theta} - L = \hat{\theta}^*_{1-\alpha/2} - \hat{\theta}$.

Conclusion (basic CI)

- Similarly, solving $\alpha/2 = \Pr(U < \theta)$ leads to $\hat{\theta} - U = \hat{\theta}^*_{\alpha/2} - \hat{\theta}$.

Basic CI (final form)

$$(L, U) = \left(2\hat{\theta} - \hat{\theta}^*_{1-\alpha/2},\ 2\hat{\theta} - \hat{\theta}^*_{\alpha/2}\right).$$

Percentile bootstrap confidence interval

- A bootstrap percentile interval uses the empirical distribution of the bootstrap replicates as the reference distribution.
- The quantiles of this empirical distribution estimate the quantiles of the sampling distribution of $\hat{\theta}$.
- This can match the true distribution better when the distribution of $\hat{\theta}$ is not normal.

Percentile CI

Let $\hat{\theta}^{*(1)}, \ldots, \hat{\theta}^{*(B)}$ be bootstrap replicates of $\hat{\theta}$. Compute the $\alpha/2$ quantile $\hat{\theta}^*_{\alpha/2}$ and the $(1-\alpha/2)$ quantile $\hat{\theta}^*_{1-\alpha/2}$ from the ecdf of the replicates. The percentile interval is

$$\left(\hat{\theta}^*_{\alpha/2},\ \hat{\theta}^*_{1-\alpha/2}\right).$$

Example: confidence intervals for patch ratio statistic

- Compute 95% confidence intervals for the ratio statistic in the patch data:

$$\hat{\theta} = \frac{\bar{y}}{\bar{z}}.$$

- Use $B = 2000$ ordinary bootstrap replicates of $\hat{\theta}$.
- Compute the normal, basic, and percentile intervals from their definitions (using the bootstrap replicates).

Application: Cross Validation

Cross validation

- Cross validation is a data partitioning method for assessing model fit, prediction accuracy, or classifier performance.
- K-**fold CV**: divide data into K folds; use each fold as the test set in turn while fitting on the remaining $K-1$ folds. The model is fitted K times.
- **Leave-one-out CV** (LOOCV): special case $K=n$, each observation serves as a test point once.
  - Equivalent to the jackknife.

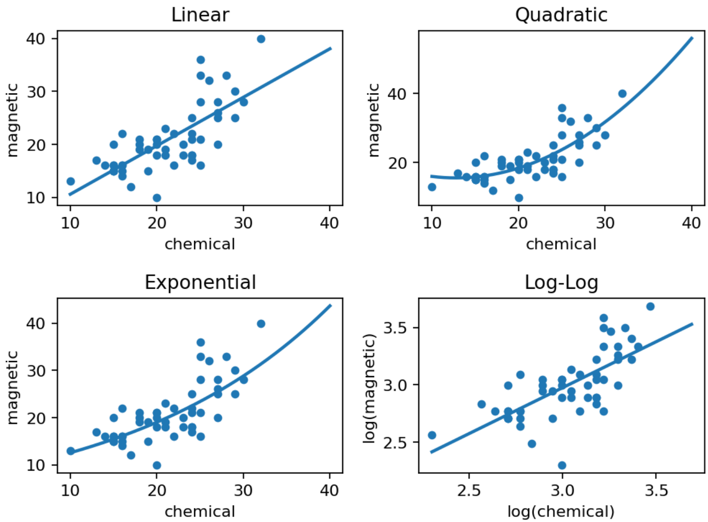
Model selection example: ironslag data

- Data: `ironslag` (DAAG), $n = 53$ measurements of iron content by two methods:
  - chemical measurement (X),
  - magnetic measurement (Y).
- A scatterplot suggests a positive association, but the relation may not be linear.

Four candidate models for predicting Y from X:

1. Linear: $Y = \theta_0 + \theta_1 X + \varepsilon$.
2. Quadratic: $Y = \theta_0 + \theta_1 X + \theta_2 X^2 + \varepsilon$.
3. Exponential: $\log(Y) = \log(\theta_0) + \theta_1 X + \varepsilon$.
4. Log-Log: $\log(Y) = \theta_0 + \theta_1 \log(X) + \varepsilon$.

Figure: four proposed models



Leave-one-out cross validation for prediction error

Procedure (n-fold leave-one-out cross validation):

1. For $k = 1, \ldots, n$, let (x_k, y_k) be the test point and fit the model on the remaining $n-1$ observations.
    1.1 Compute the predicted response $\hat{y}_k$ for the test point.
    1.2 Compute the prediction error $e_k = y_k - \hat{y}_k$.
2. Estimate prediction error:

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{k=1}^{n} e_k^2.$$

Prediction error estimates

- The following estimates for prediction error are obtained from the n-fold (leave-one-out) cross validation:

$$\big(\, 19.55644,\; 17.85248,\; 18.44188,\; 20.45424 \,\big).$$

- According to the prediction error criterion, Model 2 (quadratic) would be the best fit for the data.